ANSWER KEY & MARKING SCHEME SET A

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** north pole and enter at the south pole	Outside the magnet
2	**(b)** uniform	Parallel lines
3	**(b)** Fleming's left-hand rule	Force, field, current
4	**(a)** Correct	Two directions impossible
5	**(a)** Student A is correct	Thumb along current
6	**(c)** green	Live — red/brown
7	**(d)** the current increases	$B \propto I/r$
8	**(a)** fuse	Melts
9	**(b)** concentric circles	Centred on wire
10	**(b)** perpendicular to the field	Sin 90°

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The region around a magnet in which its magnetic force can be detected.	2
12	Hold the conductor in the right hand with the thumb along the current; curled fingers give the direction of magnetic field lines.	2
13	When the live wire and neutral wire come in direct contact, causing a sudden large current.	2
14	True — outside the magnet they go from N to S and inside from S to N.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Stretch thumb, forefinger and middle finger mutually perpendicular: forefinger — field, middle finger — current, thumb — force (motion); depends on current and strength of field (and length)	2+1
16	Field lines are concentric circles near the wire; at the centre they become straight lines perpendicular to the loop's plane; field increases with current and number of turns	3
17	Coil of many circular turns of insulated wire; field like a bar magnet, uniform inside; a soft iron core placed inside makes an electromagnet	1+1+1
	OR: Provides a low-resistance path for leakage current to earth, preventing electric shock; green wire	2+1
18	Too many appliances drawing a large current; using fuses/MCBs; not connecting many appliances to one socket	1+2
	OR: (i) doubles (ii) halves (iii) direction reverses	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) aluminium rod suspended between poles of a horse-shoe magnet; when current flows, the rod is displaced (ii) the force reverses (iii) Fleming's left-hand rule	3+1+1
	OR: (i) mains with live, neutral, earth; main fuse, meter, distribution box, parallel circuits (ii) 15 A for high-power appliances (geysers); 5 A for bulbs and fans (iii) earth — safety against leakage; fuse — protection from overloading and short circuits	2+1+2
20	(i) it deflects (ii) deflection reverses (iii) current produces a magnetic field (iv) concentric circles around wire (v) right-hand thumb rule	1+1+1+1+1
	OR: (i) emerge from N, enter S; closed curves; never intersect; crowded where strong (ii) closer lines mean a stronger field	4+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET B

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** small bar magnet	Points N–S
2	**(c)** south to north	Closed loops
3	**(b)** bar magnet	Similar pattern
4	**(a)** Correct	Fields add
5	**(b)** Student A is correct	Moving charge
6	**(c)** 220 V	AC mains
7	**(d)** red	Neutral is black
8	**(c)** soft iron core	Temporary magnet
9	**(b)** current	Forefinger: field
10	**(b)** tesla	T

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It is a small magnet; it experiences a force due to the magnetic field of the bar magnet.	2
12	Fuse, MCB (earthing)	1+1
13	Provides a low-resistance path for leakage current, protecting the user from shock.	2
14	False — the field at the centre is strong and nearly uniform, perpendicular to the plane of the loop.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lines from N to S curving outside; closed curves; never intersect; crowded near poles	1+2
16	Number of turns, strength of current, nature of core material	3
17	(i) perpendicular to field (ii) parallel to field; Fleming's left-hand rule	1+1+1
	OR: So it breaks the circuit at the supply side when excess current flows; alloy of low melting point (tin–lead)	2+1
18	Use proper fuses/MCBs; avoid many appliances on one socket; use good insulation; repair damaged wires	3
	OR: Straight: concentric circles, weakens with distance; solenoid: like bar magnet, uniform inside, stronger with more turns	3

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) field like a bar magnet; uniform inside (ii) place the rod inside the solenoid and pass current (iii) electromagnet: temporary, strength adjustable; permanent: retains magnetism	2+1+2
	OR: (i) thumb, forefinger, middle finger mutually perpendicular (ii) upward (towards top of page) (iii) loudspeakers, electric motor (fans)	2+2+1
20	(i) overloading (ii) excessive current flowed (iii) it can be reset and reacts quickly (iv) do not use many high-power appliances at once (v) wires could overheat and catch fire	1+1+1+1+1
	OR: (i) thick wire passed vertically through cardboard with iron filings; filings form concentric circles when current flows; compass shows direction (ii) filings arrange more distinctly; field becomes stronger	4+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.